

A decision has been made on JGME-D-26-00298

From em@editorialmanager.com

To Matt A. Porter | Qompass Founder & CEO

Date Tuesday, May 19th, 2026 at 10:00

Ref.: Ms. No. JGME-D-26-00298

Generative Artificial Intelligence Use in Pediatric Residency Personal Statements
Journal of Graduate Medical Education

Dear Mr. Porter,

Thank you for your review of this manuscript. The Editor has decided not to publish this paper.

A copy of the decision letter can be found below.

You can also access your review comments and the decision letter by logging on as a Reviewer.

Thank you again for your efforts, and we hope to contact you for future reviews.

To: *****

From: "" jgme@acgme.org

Subject: Your Submission

Ref.: Ms. No. JGME-D-26-00298

Generative Artificial Intelligence Use in Pediatric Residency Personal Statements
Journal of Graduate Medical Education

Dear ***** ,

Thank you very much for submitting your manuscript "Generative Artificial Intelligence Use in Pediatric Residency Personal Statements" for review. The editors appreciated the attention to an important topic. However, space constraints do not allow us to publish many of the excellent submissions that we receive. While we are unable to publish this manuscript, we appreciate the opportunity to consider your submission. Comments from our editors regarding the manuscript decision are detailed below.

Thank you again for thinking of JGME. We will look forward to further contributions from you and your colleagues.

Sincerely,

Dr. Winfred Frazier
Associate Editor

Gail M Sullivan, MD, MPH

Editor-in-Chief

Journal of Graduate Medical Education

Comments from the Editors and Reviewers:

Reviewer #1: General Comments to Authors

This manuscript addresses a timely question regarding generative AI use in pediatric residency personal statements, and the authors are commended for assembling a large, multi-year dataset. However, significant methodological concerns preclude publication in its current form. Central to this review's concerns is the broader context in which this research sits: program directors are increasingly being positioned as both consumers and enforcers of AI detection, yet the tools available to them vary enormously in validated accuracy, institutional support, and ethical safeguards. The manuscript misses a critical opportunity to engage meaningfully with this landscape, and this gap is particularly difficult to reconcile given the public availability of purpose-built GME infrastructure at the time of this manuscript's preparation. Most notably, in November 2024 the AAMC and Thalamus formally announced the expansion of Cortex, an AI/ML-enabled holistic application review platform, to all ERAS-participating programs as a complimentary benefit beginning July 2025 for the 2026 ERAS season. This announcement predates this manuscript's submission. The AAMC has confirmed on its own institutional FAQ page (<https://www.aamc.org/services/eras-institutions/faqs-aamc-thalamus>) that Cortex uses artificial intelligence, machine learning, natural language processing, and optical character recognition to aggregate key application data in alignment with the AAMC's published principles for responsible AI in residency and fellowship selection. Cortex has been operational in some capacity since 2020 and has been used to screen over three million residency and fellowship applications at institutions throughout the United States. It was piloted in the 2025 ERAS season and was fully integrated and available to all programs by July 2025. A manuscript that aims to inform program director practice on AI use in personal statement review, submitted after the AAMC's formal announcement of this tool's universal availability, and without any acknowledgment of its existence, presents a gap in the literature review that is difficult to reconcile and should be addressed in revision. Equally important is the manuscript's failure to account for the AI tools most accessible to the very applicants under study. Microsoft Copilot is now natively embedded in Microsoft 365, including Microsoft Word, the most common tool used to draft personal statements, and is physically integrated into an increasing number of student-facing devices via dedicated Copilot hardware keys. This represents arguably the most widely available and frictionless AI writing assistant available to applicants today, yet it receives no acknowledgment in the Introduction, Methods, or Discussion. A study examining AI use in personal statements that does not account for the most ubiquitous AI writing environment available to students has a significant blind spot that must be addressed.

The specific comments below detail each concern with reference to the relevant pages and line numbers of the manuscript.

Specific Comments to Authors

Comment 1

Page 1, Lines 9-12 and Page 3, Lines 9-12

Although OpenAI's ChatGPT is arguably the most well-known AI tool across the medical education landscape, it is unspecific at best and misleading at worst to equivocate ChatGPT as a catch-all for AI. The abstract and

introduction use "AI (i.e. ChatGPT)" as though these terms are interchangeable, which they are not. ChatGPT is one product from one company; AI is a broad field encompassing numerous tools, architectures, and use cases. This conflation risks biasing the reader's interpretation of the findings, misrepresenting the landscape of tools available to applicants, and limiting the generalizability of the study's conclusions. The authors should distinguish clearly between generative AI as a category and ChatGPT as a specific product throughout the manuscript.

Comment 2

Page 4, Line 14

OpenAI's GPT-4 model was released in March 2023, not 2022 as the authors have written. The factual release date for GPT-4 can be confirmed via OpenAI's own release announcement at <https://openai.com/index/gpt-4-research/>. The platform publicly released in November 2022 was ChatGPT, which at launch was powered by GPT-3.5, not GPT-4. This is not a minor distinction: the study's entire temporal framing depends on accurate characterization of which AI model generation was available during each application cycle. The authors should correct this error and clarify which model generation they believe applicants were using in each cycle year.

Comment 3

Page 6, Lines 19-21

The authors state that Quillbot and ZeroGPT were selected in part because they are "free, open-access platforms" with "good accuracy." While the intent to minimize cost barriers is understandable, this justification warrants further scrutiny on two grounds: the characterization of these platforms as free, and the adequacy of the accuracy data provided. Both Quillbot and ZeroGPT operate freemium models. While basic functionality is available at no cost, each platform offers paid tiers with expanded features and higher usage limits. The claim that these tools were selected because they are "free" may therefore be imprecise and should be clarified. More critically, the cost justification fails to account for purpose-built, validated GME infrastructure that is already available to all residency programs at no cost. The AAMC has confirmed that starting July 2025, all ERAS-participating programs receive complimentary access to the Thalamus Cortex platform (<https://www.thalamusgme.com/add-ons/cortex-application-screening>), which uses AI/ML, NLP, and optical character recognition to analyze application materials including personal statements, in alignment with the AAMC's published principles for responsible AI in residency selection (<https://www.aamc.org/services/eras-institutions/faqs-aamc-thalamus>). Thalamus publishes methodology white papers for each AI feature through its AI Hub (<https://www.thalamusgme.com/ai-hub>), including a published methodology for the Academic Career Interest Badge, which uses a large language model to analyze personal statements for expressed and implied applicant characteristics (<https://www.thalamusgme.com/blogs/methodology-for-the-academic-career-interest-badge-in-cortex>). An independent peer-reviewed early evaluation of Cortex has been published in Laryngoscope (Erickson et al., 2026, DOI: 10.1002/lary.70386), providing a citable, independently assessed evidence base that the authors could have engaged with. While Cortex is not designed as a standalone AI-generation detector, it represents the standard of purpose-built, bias-conscious, and institutionally validated tooling available to the GME community, the existence of which the manuscript does not acknowledge at all. Additionally, the authors' institution (Yale School of Medicine) provides its community with free institutional access to several enterprise-grade AI platforms, including Anthropic's Claude, Google Gemini, and Microsoft Copilot, through Yale's AI Tools and Resources portal (<https://ai.yale.edu/yales-ai-tools-and-resources>). Yale's own AI research infrastructure, including the Aneja Lab (<https://github.com/Aneja-Lab-Yale>) and its contributions to open-source medical AI such as Meditron via the Yale-EPFL

collaboration (<https://ai.meta.com/blog/llama-2-3-meditron-yale-medicine-epfl-open-source-llm/>), further underscores the breadth of validated AI tooling available to the authors' institution that was not considered in this study. The rationale for selecting consumer-grade freemium tools over these validated, institutionally supported, and purpose-built alternatives should be explicitly addressed in a revised Methods section.

Comment 4

Page 6, Lines 19-21

The claim that these platforms have "good accuracy" lacks the specificity required given the high-stakes nature of their application. Weber-Wulff et al., the very study cited as Reference 24 in this manuscript, evaluated 14 AI detection tools and found that "all scored below 80% accuracy and only 5 achieved over 70%," concluding that available detection tools are "neither accurate nor reliable" with a systematic bias toward classifying AI-generated output as human-written. Using the binary inclusive evaluation method, ZeroGPT achieved an accuracy of 74%, and only Turnitin reached the maximum of 81% across all tools tested. Critically, accuracy dropped sharply for hybrid writing scenarios: tools achieved only approximately 50% accuracy for AI-generated text that was manually edited, and approximately 26% accuracy for AI-generated text that had been machine-paraphrased, the very conditions most likely to appear in real residency personal statements. The paper explicitly warns that "reports provided by detection tools for AI-generated text cannot be used as the only basis for reporting students for cheating." In the context of residency application screening, where a false positive could unjustly stigmatize an applicant's candidacy, the phrase "good accuracy" is insufficient and unsupported even by the authors' own cited literature. The authors should directly acknowledge these limitations and the specific performance data of ZeroGPT and Quillbot as documented in Reference 24, and address whether a consensus-based approach aggregating multiple detectors was considered as part of the study design.

Comment 5

Page 6, Lines 19-23; Page 7, Lines 24-26; Figure 2, Page 21

The study reports AI detection as a continuous variable (mean AI%) throughout the analysis, yet Figure 2 and the associated results text present a binary outcome defined as "any amount of AI detection," a threshold of greater than 0%. This creates two distinct analytical problems. First, using greater than 0% as the threshold for "AI use" is methodologically unjustifiable given the known baseline false positive rates of both platforms; as the authors themselves acknowledge, 2022 statements written prior to the public release of ChatGPT were nonetheless flagged as containing AI-generated content, with a mean Quillbot score of 4.9% (SD 12.9) and ZeroGPT mean of 35.9% (SD 21.1) (Table 2, page 20). Any threshold that classifies these pre-ChatGPT statements as AI-generated is, by definition, capturing noise rather than signal. Second, the continuous mean AI% data presented in Table 2 treat a statement scored at 5% identically to one scored at 95% in group-level comparisons, despite these scores having fundamentally different clinical and operational interpretability. The authors should: (1) define and justify a meaningful minimum threshold for classifying a statement as AI-assisted (e.g., 20% or 50%), informed by the false positive rates documented in their own baseline 2022 data; (2) reanalyze and represent Figure 2 using this validated threshold; and (3) consider stratified analyses across AI% bands (e.g., 1-20%, 21-50%, greater than 50%) to provide findings that are actionable for residency program directors.

Comment 6

Page 6, Lines 28-32; Page 7, Lines 19-22; Table 2, Page 20; Page 8, Lines 1-5

The authors state that applicants in the 2022 cycle are presumed not to have had access to ChatGPT, given that the platform was not publicly released until November 30, 2022, after the ERAS application deadline for that cycle had passed. This presumption is reasonable and effectively renders the 2022 cohort a natural negative control: any AI-generated content detected in 2022 personal statements cannot reflect actual applicant AI use and must therefore represent false positives. Table 2 reveals that both platforms produced substantial detection scores in this negative control year: Quillbot returned a mean of 4.9% (SD 12.9) and ZeroGPT returned a mean of 35.9% (SD 21.1) for 2022 applicants. The ZeroGPT figure is particularly concerning. A mean false positive rate of 35.9% on a cohort that could not have used AI represents a profound baseline noise level that fundamentally compromises the interpretability of all subsequent year-over-year comparisons. The authors acknowledge this briefly in the Discussion, noting that "both detectors produced false positives," but do not adequately engage with its analytical implications. Specifically, the authors must address the following: (1) If ZeroGPT flags an average of 35.9% AI content in verified human-written 2022 statements, on what basis can any ZeroGPT score in 2023 or 2024 be interpreted as evidence of AI use rather than detector noise? (2) The year-over-year increase reported as a central finding of this study, from a ZeroGPT mean of 35.9% in 2022 to 51.5% in 2023 and 65.3% in 2024, must be interpreted against this baseline, yet no such adjustment or sensitivity analysis is performed. (3) Figure 2, which reports "any AI detection" as a binary outcome, is rendered uninterpretable if the baseline false positive rate already produces near-universal detection in human-written text. The authors should reframe their trend analysis using the 2022 false positive rate as a noise floor, conduct a sensitivity analysis excluding scores below the established baseline threshold, and revise their conclusions accordingly. Failure to do so means the manuscript's primary finding, that AI use is increasing, cannot be distinguished from the alternative explanation that detector sensitivity or bias increased over time.

Comment 7

Page 13, Lines 1-4 (Reference 11)

Angus et al.'s manuscript "Internal Medicine Residency Program Directors' Screening Practices and Perceptions About Recruitment Challenges" carries a potential citation date discrepancy. Depending on the citation convention used, this reference may reflect the online-ahead-of-print publication date of 2020 or the print issue date of 2021. The discrepancy was identified by visiting the DOI hyperlink provided by the authors (<https://doi.org/10.1097/ACM.0000000000003086>). The authors should verify and standardize their citation date convention consistently across all references in the manuscript. If following standard print issue citation convention, the reference should read: Acad Med. 2021 Apr;95(4):582-589. If following online-ahead-of-print convention, the reference should read: Acad Med. 2020;95(4):582-589.

Reviewer #2: Thank you for the opportunity to review this manuscript. The authors present an analysis of pediatric residency applicants' personal statement for generative AI (GenAI) use. The topic is timely and of interest to JGME readers, though not novel and the results are not surprising. The most significant issue with the study is the simultaneous conclusion that AI-detector tools are flawed, and that AI use (as measured by these AI-detection tools) is increasing. It is difficult to reconcile these; it is akin to saying that "a new test to detect COVID has poor test performance, and when we used this test we found that COVID incidence is increasing." Several other concerns are enumerated below:

Abstract

1. Page 2 Line 9: This line seems to suggest that ChatGPT is the only AI tool. Suggest "e.g." instead of "i.e."
2. Line 9: It is more accurate to describe ChatGPT as a "Large Language Model" or even "Generative AI" - suggest change this throughout rather than the broader marketing term "AI" that indicates a much higher level of performance than the current tools.
3. Line 43: "No difference existed" - this is more accurately written as "we were unable to detect a difference"
4. Line 50: See concerns in the paragraph above re: clashing conclusions - it does not seem right to conclude that "AI use is increasing" and at the same time "the tools we used to measure AI use are flawed"

Introduction

5. Page 3 Line 6: "is raising concerns" - needs reference
6. Line 33: Please also reference COMLEX scores when talking about the generic applicant
7. Line 41: Delete "objective" - there are several concerns with the "objectivity" of these measures
8. Page 4 Line 10: "i.e." - as above #1
9. Line 17: This, and several other examples, seem to suggest that there has been plenty of literature showing AI use is increasing in personal statements. The authors should try to illustrate the knowledge gap why this question is particularly important to study in pediatrics, for example why they think the average pediatric applicant is different from the other specialties where this has already been described.

Methods

10. Page 5 line 7: Suggest "Connecticut" rather than "New England" as the latter may not be familiar to an international audience
11. Line 9: ERAS has previously been defined as an abbreviation
12. Line 12: While it appears that consent was waived, it's concerning that applicants' original creative work may have been fed into a third-party engine without their consent. The authors should describe what was done to ensure that the applicants' personal statements were not ingested by these tools or used for commercial purposes by the companies. It could be argued that applicants' personal statement use in this manner violates ERAS restrictions on commercial use of data.

Results

13. Page 6 Line 46: "no statistically significant differences" - should be stated as "we were unable to detect any statistically significant differences" (type 2 error)

Discussion

14. Page 7 Line 7: There is a lot of evidence published in technology literature about how the AI detection tools are greatly flawed. Use of these tools to make decisions, or for published research, is problematic. Some of the concerns have been highlighted by the authors (poor correlation, false positives and false negatives). In this study, these concerns seem almost like an afterthought, as the main conclusions power ahead with statements like "AI use is increasing" which are based on the flawed tools that this paragraph tries to discuss.
15. Line 38: Several results are introduced in this paragraph without being first described in the results section. Recommend disclosing the results earlier, then presenting the authors' opinion in this section.
16. Line 45: "higher means of AI-detection" is unclear - perhaps "higher percentages of AI text"?

Conclusions

17. Page 10 Line 24: See #4 above.

Figures

18. Please ensure that appropriate copyright releases have been obtained from the two commercial tools, before reproducing their output

19. Figure 2 seems to indicate that ZeroGPT found almost 100% AI use every year?

Reviewer #3: Review of Generative Artificial Intelligence Use in Pediatric Residency Personal Statements

I do think this article is interesting, however it stands out in that it does show that the AI detection tools are unreliable.

1) Subillaga O, Coulter AP, Tashjian D, Seymour N, Hubbs D. Artificial Intelligence-Assisted Narratives: Analysis of Surgical Residency Personal Statements. J Surg Educ. 2025 Dec;82(12):103566. doi: 10.1016/j.jsurg.2025.103566. Epub 2025 Jun 18. PMID: 40537315.

2) Patel Housley P, Fadhel-Hernandez V, Montanez-Wiscovich M, Neal D, Bohannon M, Schiefer MA. Rising Influence of Artificial Intelligence: Trends in Large Language Model Usage in Dermatology Residency Personal Statements. J Drugs Dermatol. 2025 May 1;24(5):544-547. doi: 10.36849/JDD.8914. PMID: 40327572.

These papers are quite alike since they come from a single center on the east coast. While they cover different specialties, their findings aren't particularly new, and I don't think they contribute much beyond confirming what is already commonly understood.

It would be valuable to examine the percentage of AI-generated content in personal statements among matched applicants. Furthermore, it is important to consider whether the information presented in these statements aligns with the actual characteristics of the individuals. Students have always had the option to embellish/falsify their personal statements; the advent of AI does not fundamentally alter this possibility. Additionally, a personal statement may often reflect a student's writing ability more than it reveals their true self. For example, students who are less skilled writers, but excellent pediatricians may benefit from AI-assisted applications. This raises pertinent questions: Does AI enable candidates to showcase their strengths by reducing the cognitive load associated with writing? Ultimately, does this impact outcomes? What are the results for learners admitted into your program? These questions would be more valuable than asking if AI use increases and can we reliably detect it, because as the end of the day does it matter if someone used Chat GPT to write their personal statement?

I have included a few areas in which I think you could take your paper one step further to really begin to answer down stream questions, given the data that you have already.

Page 3 Line 28: Importance for what? In the rest of your paragraph, you discuss importance of initial screening for interview offers. I would say this in the intro sentence, such as: "Although program directors debate the usefulness of personal statements, many still consider them important for selecting residency interview candidates."

Page 7 Line 47: I am nervous about the implications of this portion of the discussion. I agree that we cannot draw conclusions about this result given that it could be the bias of the baseline tool used to detect AI. However, downstream questions could be, did the number of Middle Eastern or "other" applicants that were interviewed during these years increase compared to years prior to the use of AI? Did this tool help them to be more successful in reaching an interview? If so, did the diversity of your incoming classes change because of this? Were these applicants successful residents and physicians?

Page 8 line 32: Does using AI for personal statements even matter? Besides the opinion of attendings, does the performance of the applicant, the type of physician they are or how well they can be trained change? PDs are concerned about work ethic, but is this true, this is all based on opinions with very little performance data. Could you point this out?

Page 9 Line 22: This is one of the most important paragraphs in the study. As it really does show that this study is

essentially a negative study. This study did not work, 2 different detectors gave two different answers and the truth is unknown. I think we should be sharing these studies, but I think this paper needs to be re-written to emphasize this point. Currently, the abstract reads as though there are measurable results.

Page 10 Conclusion: What is in the conclusion is not supported by the data. Having applicants disclose that they use AI is not studied in this paper and should not be a conclusion. We have never required students who had their personal statement read by mentors as something that needs to be disclosed. Never required those very lucky students who have the means to hire a company to help them with their applications to disclose. Now, we have a tool that can be equitably used and we are proposing that this should be disclosed? Where is this part of the conclusion supported? The conclusions from this paper should be limited to the final sentence: Given that current AI detectors remain unreliable and risk introducing bias into the residency selection process, PDs should use them cautiously.

AE comments:

Thank you for submitting this article.

This manuscript addresses a timely topic, but further clarification of the central research question would strengthen its overall impact. At present, it is difficult to determine whether the primary concern is ethical (e.g., appropriate use of AI), related to the authenticity and accuracy of personal statements, or focused on implications for residency selection processes. Explicitly stating this framing early in the manuscript would improve coherence and guide the reader more effectively.

In the introduction, I recommend revising the language to better reflect the target population; for example, replacing “general public” with medical students or residency applicants. Additionally, the characterization of personal statements as “integral” may be overstated. Their importance has varied over time and across programs, and while they may play a role in holistic review, some selection processes increasingly rely on other metrics, such as USMLE Step 2 CK scores in the post–Step 1 pass/fail era. The discussion of applicants using paid services to revise personal statements may also overestimate the prevalence of this practice.

The manuscript would benefit from a more robust discussion of bias in personal statement evaluation, particularly the concept of “goodness of fit.” This construct is often unstructured and may reinforce similarity bias rather than identifying candidates who bring diverse perspectives or strengths. Given the study topic, incorporating how AI might interact with or obscure these biases would add important depth and relevance.

In the methods section, the rationale for including demographic variables such as gender, self-identity, and medical school should be more clearly justified. It would be helpful to explicitly state how these variables relate to the study aims or hypotheses. In the discussion, I encourage the authors to further address the limitations of AI detection, including whether alternative tools were considered and the broader challenges in distinguishing AI-generated from AI-assisted content. The section discussing Middle Eastern students feels overly specific and somewhat disconnected from the main argument; unless strongly supported by the data and central to the study question, this section may benefit from revision or removal.

Finally, the conclusions should be closely aligned with the data and avoid overstating implications, particularly given the methodological limitations. The inclusion of the demographic table is also unclear in its current form; the authors should clarify what conclusions are intended from this information or consider whether it adds meaningful value to the manuscript.

Overall, refining the study’s focus, strengthening the discussion of bias and selection processes, and ensuring alignment between the data and conclusions would substantially improve the manuscript.

Kind regards,

Gail M Sullivan, MD, MPH

Editor-in-Chief

Journal of Graduate Medical Education

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